

SIERRA VISTA BLACK TOWER, AZ, USA

WMO: 722813

Lat: 31.607N

Lon: 110.428W

Elev: 4768

StdP: 12.33

Time Zone: -7.00 (W07)

Period: 08-19

WBAN: 53146

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.4	(c) 29.7	(d) -4.4	(e) 5.2	(f) 60.3	(g) -0.9	(h) 6.3	(i) 61.6	(j) 30.2	(k) 53.1	(l) 26.9	(m) 52.3	(n) 7.3	(o) 230	(p) 0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 27.8	(c) 95.4	(d) 59.6	(e) 93.2	(f) 59.6	(g) 91.2	(h) 59.7	(i) 67.7	(j) 80.5	(k) 66.8	(l) 79.8	(m) 66.1	(n) 79.1	(o) 12.3	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	108.6	70.5	63.4	104.5	70.3	62.4	100.8	70.5	34.8	80.1	34.1	79.8	33.4	78.7	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.8	24.6	20.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	18.4	70.7	2.8	2.2	16.4	72.3	14.8	73.6	13.2	74.9	11.2	76.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	47.5	50.5	56.4	61.2	67.4	78.3	77.4	76.3	72.5	65.0	55.8	48.1		
	DBStd	12.42	6.92	7.45	6.61	6.17	6.07	4.28	3.60	3.43	3.75	5.64	6.62	7.07		
	HDD50	377	127	77	22	6	0	0	0	0	0	2	24	119		
	HDD65	2276	543	407	273	138	47	0	0	0	2	66	277	523		
	CDD50	5159	49	91	219	343	540	848	851	815	674	469	199	61		
	CDD65	1583	0	1	6	25	121	398	386	350	226	68	2	0		
	CDH74	17742	0	39	197	559	1813	5319	3654	3165	2042	855	98	1		
	CDH80	7485	0	1	15	94	649	2859	1628	1338	717	174	10	0		
(14) Wind	WSAvg	9.7	8.8	10.3	10.9	11.7	11.5	10.6	8.8	7.9	8.1	9.0	9.3	9.2		
(15) Precipitation	PrecAvg	13.3	1.0	0.6	0.5	0.3	0.2	0.6	3.5	3.4	1.5	0.6	0.5	0.7		
	PrecMax	19.8	6.0	1.7	2.5	1.5	1.5	3.7	5.6	5.9	5.6	6.8	3.1	3.0		
	PrecMin	8.1	0.0	0.0	0.0	0.0	0.0	0.0	1.5	1.5	0.5	0.0	0.0	0.0		
	PrecStd	3.6	1.3	0.6	0.6	0.4	0.3	0.8	1.3	1.1	1.0	1.3	0.6	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	78.4	82.1	85.4	92.0	100.7	96.4	94.9	91.5	87.4	81.1	73.3		
		MCWB	47.0	49.3	50.2	51.4	54.7	59.1	62.2	63.8	61.1	56.8	50.4	46.9		
	2%	DB	68.3	74.6	79.1	82.8	89.3	97.1	93.6	92.1	89.3	84.3	76.8	69.4		
		MCWB	45.5	47.3	49.0	50.5	53.7	58.3	62.6	63.6	60.9	56.1	50.6	46.3		
	5%	DB	65.1	70.1	76.1	80.2	86.8	94.7	91.3	89.8	86.9	81.7	73.5	65.7		
		MCWB	44.3	45.6	48.0	49.7	52.6	57.8	62.8	63.6	60.9	55.9	49.7	45.9		
	10%	DB	61.2	65.4	72.5	77.2	83.9	92.6	88.7	87.4	84.5	79.1	70.2	61.3		
		MCWB	43.0	44.1	46.8	48.7	51.7	57.4	62.8	63.6	60.6	54.7	48.6	43.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	52.0	51.7	53.9	57.2	65.5	68.5	69.2	67.3	62.2	55.5	52.2		
		MCDB	65.2	62.8	74.4	77.8	83.0	80.4	80.0	82.8	79.0	76.2	68.8	59.9		
	2%	WB	48.6	50.2	50.3	52.0	55.3	63.9	67.5	68.1	66.0	60.3	53.1	49.5		
		MCDB	58.9	63.5	72.9	77.2	80.9	82.0	79.1	81.7	77.9	75.1	68.1	62.1		
	5%	WB	46.6	48.2	49.1	50.7	53.9	62.5	66.6	67.2	64.8	58.9	51.4	47.4		
		MCDB	59.5	63.8	71.7	76.1	81.6	82.3	79.0	80.9	76.9	74.4	68.8	62.2		
	10%	WB	44.6	46.2	47.8	49.6	52.6	61.2	66.0	66.4	63.8	57.2	50.0	45.6		
		MCDB	58.5	63.2	70.1	74.7	80.7	82.5	78.9	79.8	76.1	73.4	67.1	59.2		
(35) Mean Daily Temperature Range	MDBR		24.1	24.7	26.7	27.6	28.0	27.8	21.4	22.0	23.1	25.8	25.0	22.5		
	5% DB	MCDBR	28.0	31.4	31.1	30.6	30.6	29.6	24.9	25.1	26.1	28.1	28.2	27.1		
		MCWBR	13.5	14.0	12.7	12.7	11.6	10.3	6.7	7.3	8.5	11.1	12.3	13.0		
	5% WB	MCDBR	22.3	25.0	28.0	28.1	27.9	24.8	20.1	20.9	20.3	23.8	24.4	22.9		
		MCWBR	11.6	11.7	11.8	12.1	10.7	7.5	5.8	6.9	7.2	9.8	11.3	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.247	0.262	0.271	0.285	0.310	0.362	0.349	0.313	0.278	0.253	0.242		
	taud		2.688	2.681	2.603	2.562	2.558	2.493	2.469	2.494	2.576	2.626	2.657	2.679		
	Ebn at Noon		321	328	326	324	317	307	291	295	304	311	314	316		
	Edh at Noon		23	25	30	32	32	35	35	34	30	26	23	22		
(44) All-Sky Solar Radiation	RadAvg		1176	1473	1944	2376	2604	2539	2041	1980	1846	1618	1295	1080		
	RadStd		80	119	108	82	102	149	145	78	92	102	64	49		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A

Nomenclature: See separate page